# Elementary Financial Analysis

## Financial Statements

# Types of Ownership & Business Organizations

There are three basic ways to structure a business organization.

A **sole proprietorship** (sole trader, single proprietor) is a business **owned by one person**. It is the **simplest and least regulated** form of business to start (in essence, all you need is a business name), and accounts for the largest number of businesses. Under a sole proprietorship, the **owner keeps all the profits, but at the same time, has *unlimited liability***; that is, the owner is personally responsible for all business debts and the creditors can come after even his or her personal assets in order to recoup debts.

A **partnership** is a business **owned by two or more people (partners)**. In a **general partnership**, the partners run the business together and share all profits and losses (unlimited liability) according to a partnership agreement. In a ***limited partnership*, some partners are involved only as investors (limited partners**) and they let one or more general partners take charge of day-to-day operation. The limited partners have *limited liability*; that is, they are only liable for up to the amount of their investment and their **personal assets are protected** from the creditors of business debts.

# Types of Ownership & Business Organizations

A corporation is owned by shareholders. The shareholders elect the board of directors, and the board is responsible for selecting the managers to run the business in the interest of shareholders. A corporation is set up as a business entity with its own rights and responsibilities, separate from the owners. This means that the corporation is responsible for its own debts, and the owners have limited liability (up to the amount of their investment).

The main advantage of incorporating a business is its relative ease in raising capital for a growth opportunity. Sole proprietorships and partnerships, although easier to start up than corporations, are both disadvantaged by personal liability issues, limited availability of equity, and difficulty of ownership transfer, all of which contribute to potential difficulties in

# Why Engineers should learn Financial Management

With the growth in importance of small technology-based enterprises, many engineers have taken on broad managerial responsibilities that include financial accounting. The object of financial accounting is to provide information to internal management and interested external parties. **Internally, management uses financial accounting information for processes such as budgeting, cash management, and management of long-term debt. External users include actual and potential investors and creditors who wish to make rational decisions about an enterprise. External users also include government agencies concerned with taxes and regulation. Areas of interest to all these groups include an enterprise's revenues and expenses, and assets (resources held by the enterprise) and liabilities (claims on**

# Assets, liabilities, equity, revenue and expense

- *Assets*: are **resources owned by a business**. The assets should provide future service or benefit to the business such as furniture, car, cash, office supplies and so on.
- *Liabilities*: are **creditors claim on total assets** of the business - Firm's existing debts and obligations. Example: account payables, wages payable, interest payable and so on.
- *Owner's equity*: it is the **ownership claims on total assets**. It equal to total assets minus total liabilities. In sole proprietor ship and partnership form of business, the principal sub divisions of owner's equity are; *capital, drawings, revenues and expense.*
- *Capital*: is the term used to describe the **owner's investment in the business** an investment made increases capital, ownership claims on the total asset. Example: Mr. X capital, ABC capital and so on.
- *Drawing*: refers to **owners withdraw cash or other assets from the business for personal use**. This decreases owner's equity in the business.
- *Revenue*: is the **gross increase in owner's equity resulting from business activities** for instance sales revenue, fees revenue, rent revenue and so on. revenue s total earning
- *Expense*: is the **cost of assets consumed or services used in the process of earning revenue** such as cost of goods sold, wages expense, utility expense etc. It decreases owner's equity.

# Basic Accounting Equation

Every business transaction, no matter how simple or complex, can be expressed in terms of its effect on the accounting equation. Regardless of whether a business grows or contracts, the equality between the assets and the claims against the assets is always maintained.

$$\text{Owners' equity} = \text{Assets} - \text{Liabilities}$$

or

$$\text{Assets} = \text{Liabilities} + \text{Owners' equity}$$

# Basic Accounting Equation – A = L + O.EQ

Assume Mr. X establishes sole proprietorship business that operates taxi services in Dessie town known as **ABC Taxi Company** on January 2016.

a) In order to start up the business, Mr. X deposited $10,000 in a bank account in the name of ABC Taxi Company.

b) Mr. X purchased land, which costs $7,500 by paying cash, to be used as a future site for the business.

c) Mr. X purchased $850 of gasoline, oil, and other supplies and agreed to pay the money in the near future.

d) Mr. X paid for creditor $400 for the supplies purchases in the above transaction.

e) Mr. X taxi provided taxi services for the customers and earned fares (revenue) of $ 4,500, Mr. X received the total amount of cash from the customers.

f) Mr. X taxi incurred and paid the following expenses during the month: wages $1,125; rent $850; utilities $ 150; miscellaneous $ 75.

g) Mr. X’s supplies on hand (not yet used) accounts for $ 250 at the end of the month, the reminder 600 (850 - 250) have been used in the operation of the business.

h) At the end of the month, Mr. X withdraws $1,000 in cash from the business for his personal use.

# Basic Accounting Equation

| Assets | | Liabilities | | Owner's Equity | |
|---|---|---|---|---|---|
| Cash | 10000 | | . | Owner invested | 10000 |
| Land | 7500 | | . | | . |
| Cash | -7500 | | . | | . |
| Gasoline, oil & supplies | 850 | Account's Payable | 850 | | . |
| Cash | -400 | Account's Payable | -400 | | . |
| Cash | 4500 | | . | Fees earned | 4500 |
| Cash | -2200 | | . | Expense incurred | -2200 |
| Supplies | -600 | | . | Supplies Used/Expense | -600 |
| Cash | -1000 | | . | Owner's withdrawal | -1000 |
| | 11150 | | 450 | | 10700 |
| 11150 | | 11150 | | | |

# Measuring firm's performance - Financial Statements

The Performance of an organization is measured with the income statement and balance sheet.

The income statement summarizes revenues and expenses over a period of time. It is prepared by listing the revenues earned during a period and the expenses incurred during the same period, and by subtracting total expenses from total revenues, arriving at a net income. An income statement is always associated with a particular period of time, be it a month, quarter, or year.

The balance sheet, in contrast to the income statement, is a snapshot of the financial position of a firm at a particular point in time. The financial position is summarized by listing the assets of the firm, its liabilities (debts), and the equity of the

# Components of Balance Sheet – Assets & Classification

**Current assets** are **cash and other assets that could be converted to cash within a relatively short period of time,** usually a year or less. Inventory and accounts receivable are examples of non-cash current assets.

non current: they can be converted to cash but are not cash yet

# Components of Balance Sheet – Current Assets & Placement

Liquidity of an asset refers to the length of time it takes to convert them to cash. Assets are arranged in order of liquidity. The most liquid assets appear at the top of the page, the least liquid assets at the bottom of the page.

1. **Cash** is the most liquid of all assets, it is always listed first. Current assets are so critical that they are separately broken out and totalled. They are what will hold the business afloat for the next year. **Cash-equivalent items** are also listed and include marketable securities and short-term investments.
2. The second account is ***accounts receivable***—money that is owed the firm, but that has not yet been received.
3. The third account is ***inventories***, which show the dollar amount that Dell has invested in raw materials, work in process, and finished goods available for sale.

# Components of Balance Sheet – Long Term /Fixed Assets

**Fixed assets** are **relatively permanent and take time to convert into cash.** The most common fixed asset is the physical investment in the business, such as land, buildings, factory machinery, office equipment, and automobiles.

With the exception of land, most fixed assets have a limited useful life. For example, buildings and equipment are used up over a period of years. Each year, a portion of the usefulness of these assets expires, and a portion of their total cost should be recognized as a depreciation expense. The term *depreciation* denotes the accounting process for this gradual conversion of fixed assets into expenses. *Property, plant and equipment, net* thus represents the current book value of these assets after deducting depreciation expenses.

# Other Assets

Finally, **other assets** include **investments made in other companies** and **intangible assets such as goodwill, copyrights, franchises**, and so forth. Goodwill appears on the balance sheet only when an operating business is purchased in its entirety. Goodwill indicates any additional amount paid for the business above the fair market value of the business.

## Liabilities and Stockholders' Equity (Owners' Net Worth)

The claims against assets are of two types: liabilities and stockholders' equity. The liabilities of a company indicate where the company obtained the funds to acquire its assets and to operate the business. Liability is money the company owes. Stockholders' equity is that portion of the assets of a company which is provided by the investors (owners). Therefore, stockholders' equity is the liability of a company to its owners.

# Liabilities and Stockholders' Equity (Owners' Net Worth)

- **Current liabilities** are those debts which must be paid in the near future (normally, within one year). The major current liabilities include accounts and notes payable within a year. Also included are accrued expenses (wages, salaries, interest, rent, taxes, etc., owed, but not yet due for payment), and advance payments and deposits from customers.
- **Other liabilities** include *long-term liabilities*, such as bonds, mortgages, and longterm notes, that are due and payable more than one year in the future.
- **Stockholders' equity** represents the amount that is available to the owners after all other debts have been paid. The stockholders' equity, or **net worth**, is a residual:

## Liabilities and Stockholders' Equity (Owners' Net Worth)

**Retained earnings** represent the **cumulative net income of the firm since its inception, less the total dividends that have been paid to stockholders.** In other words, retained earnings indicate the amount of assets that have been financed by plowing profits back into the business. Therefore, retained earnings belong to the stockholders.

# Components of Balance Sheet – Owner's Equity

**Retained earnings** includes the cumulative sum of earnings from normal operations in addition to gains (or losses) from transactions like the sale of plant assets or investments—the proceeds of which have been reinvested in the business (i.e., not paid out as dividends). Firms retain earnings mainly to expand operations through the purchase of additional assets. Contrary to what one may think, retained earnings do not represent cash. They may be invested in assets such as equipment and inventory.

# Components of Balance Sheet – Owner's Equity

The difference between a business's assets and its liabilities is the amount due to the owners—their equity in the business. That is, owners' equity is what is left over from assets after claims of others are deducted. We have, therefore,

$$\text{Owners' equity} = \text{Assets} - \text{Liabilities}$$

or

$$\text{Assets} = \text{Liabilities} + \text{Owners' equity}$$

# Balance Sheet - Basic Accounting Equation

The balance sheet gets its name from the idea that the total assets are equal in value to or *balanced by* the sum of the total liabilities and the owners' equity. A simple way of thinking about it is that the capital used to buy each asset has to come from **debt (liabilities)** and/or **equity (owners' equity).** At a company's start-up, the original shareholders may provide capital in the form of equity, and there will also likely be **debt capital**.

As the company undertakes its business activities, it will make or lose money. As it does so, assets will rise or fall and equity and/or debt will rise or fall correspondingly. For example, if the company makes a profit, the profits will either pay off debts, be invested in new assets, or be paid as dividends to shareholders.

## Major Electric Company
## Balance Sheet as of November 30, 2013

| | | |
|---|---|---|
| **Assets** | | |
| *Current Assets* | | |
| Cash | | $ 39 000 |
| Accounts receivable | | 27 000 |
| Raw materials inventory | | 52 000 |
| Finished goods inventory | | 683 000 |
| Total Current Assets | | $ 801 000 |
| *Long-Term Assets* | | |
| Equipment | $6 500 000 | |
| Less accumulated depreciation | 4 000 000 | 2 500 000 |
| Buildings | 1 750 000 | |
| Less accumulated depreciation | 150 000 | 1 600 000 |
| Land | | 500 000 |
| Total Long-Term Assets | | $ 4 600 000 |
| **Total Assets** | | **$5 401 000** |
| **Liabilities and Owners' Equity** | | |
| *Current Liabilities* | | |
| Accounts payable | | $ 15 000 |
| Loan due December 31, 2013 | | 75 000 |
| Total Current Liabilities | | 90 000 |
| *Long-Term Liabilities* | | |
| Loan due December 31, 2018 | | 1 000 000 |
| **Total Liabilities** | | **$1 090 000** |
| *Owners' Equity* | | |
| Common stock: 1 000 000 shares at $3 par value per share | | $ 3 000 000 |
| Retained earnings | | 1 311 000 |
| **Total Owners' Equity** | | **$4 311 000** |
| **Total Liabilities and Owners' Equity** | | **$5 401 000** |

Ian Claymore is the accountant at Major Electric. He has just realized that he forgot to include in the balance sheet for November 30, 2013, a government loan of $10 000 to help in the purchase of a $25 000 test stand (which he also forgot to include). The loan is due to be repaid in two years. When he revises the statement, what changes should he make?

The government loan is a long-term liability because it is due in more than one year. Consequently, an extra $10 000 would appear in long-term liabilities. This extra $10 000 must also appear as an asset for the balance to be maintained. The $25 000 value of the test stand would appear as equipment, increasing the equipment amount to $6 525 000. The $15 000 extra must come from a decrease in cash from $39 000 to $24 000.

Depending on the timing of the purchase, depreciation for the test stand might also be recognized in the balance sheet. Depreciation would reduce the net value of the equipment by the depreciation charge. The same amount would be balanced in the liabilities section by a reduction in retained earnings. ■

# Income Statement

An **income statement** summarizes an enterprise's revenues and expenses over a specified accounting period. Months, quarters, and years are commonly used as reporting periods. As with the balance sheet, the heading of an income statement gives the name of the enterprise and the reporting period. The income statement first lists revenues, by type, followed by expenses. Expenses are then subtracted from revenues to give income (or profit) before taxes. Income taxes are then deducted to obtain net income. See Close-Up 6.3 for a measure used for operating profit. The income statement summarizes the revenues and expenses of a business over a period of time. However, it does not directly give information about the generation of cash. For this reason, it may be useful to augment the income statement with a **statement of changes in financial position** (also called a cash flow statement, a statement of sources and uses of funds, or a funds statement), which shows the amounts of cash generated by a company's operation and by other sources, and the amounts of cash used for investments and other

## CLOSE-UP 6.3 Earnings Before Interest and Income Tax (EBIT)

Income before taxes *and* before interest payments—that is, total revenue minus operating expenses (all expenses except for income tax and interest)—is commonly referred to as the earnings before interest and taxes (EBIT). EBIT measures the company's operating profit, which results from making sales and controlling operating expenses. Due to its focus on operating profit, EBIT is often used to judge whether there is enough profit to recoup the cost of capital (see Appendix 4A).

**Major Electric Company**
**Income Statement**
for the Year Ended November 30, 2013

| | | |
|---|---|---|
| **Revenues** | | |
| Sales | $7 536 000 | |
| Management fees earned | 106 000 | |
| **Total Revenues** | | **$7 642 000** |
| | | |
| **Expenses** | | |
| Cost of goods sold | $6 007 000 | |
| Selling costs | 285 000 | |
| Administrative expenses | 757 000 | |
| Interest paid | 86 000 | |
| **Total Expenses** | | **$7 135 000** |
| | | |
| Income before taxes | | $507 000 |
| Taxes (at 40%) | | 202 800 |
| **Net Income** | | **$304 200** |

You have been given the following data from the Fine Fishing Factory for the year ending December 31, 2013. Construct an income statement and a balance sheet from the data. Depreciation on office equipment Rs. 850.

| | |
|---|---|
| Accounts payable | $ 27 500 |
| Accounts receivable | 32 000 |
| Advertising expense | 2500 |
| Bad debts expense | 1100 |
| Buildings, net | 14 000 |
| Cash | 45 250 |
| Common stock | 125 000 |
| Cost of goods sold | 311 250 |
| Depreciation expense, buildings | 900 |
| Government bonds | 25 000 |
| Income taxes | 9350 |
| Insurance expense | 600 |
| Interest expense | 500 |
| Inventory, December 31, 2013 | 42 000 |
| Land | 25 000 |
| Machinery, net | 3400 |
| Mortgage due May 30, 2015 | 5000 |
| Office equipment, net | 5250 |
| Office supplies expense | 2025 |
| Other expenses | 7000 |
| Total expenses | 84 500 |
| Prepaid expenses | 3000 |
| Retained earnings | ? |
| Salaries expense | 69 025 |
| Sales | 421 400 |
| Taxes payable | 2500 |
| Wages payable | 600 |

## Fine Fishing Factory
## Balance Sheet
## as of December 31, 2013

| | |
|---|---|
| **Assets** | |
| **Current assets** | |
| Cash | $ 45 250 |
| Accounts receivable | 32 000 |
| Inventory, December 31, 2013 | 42 000 |
| Prepaid expenses | 3000 |
| Total current assets | $ 122 250 |
| **Long-term assets** | |
| Land | 25 000 |
| Government bonds | 25 000 |
| Machinery, net | 3400 |
| Office equipment, net | 5250 |
| Buildings, net | 14 000 |
| Total long-term assets | $ 72 650 |
| **Total assets** | **$194 900** |
| **Liabilities** | |
| **Current liabilities** | |
| Accounts payable | $ 27 500 |
| Taxes payable | 2500 |
| Wages payable | 600 |
| Total current liabilities | $ 30 600 |
| **Long-term liabilities** | |
| Mortgage due May 30, 2015 | 5000 |
| Total long-term liabilities | $ 5000 |
| **Total liabilities** | **$ 35 600** |
| **Owners' Equity** | |
| Common stock | $ 125 000 |
| Retained earnings | 34 300 |
| **Total owners' equity** | **$ 159 300** |
| **Total liabilities and owners' equity** | **$194 900** |

## Fine Fishing Factory
## Income Statement
## for the Year Ending December 31, 2013

| | |
|---|---|
| **Revenues** | |
| Sales | $421 400 |
| Cost of goods sold | 311 250 |
| Net revenue from sales | $110 150 |
| **Expenses** | |
| Salaries expense | 69 025 |
| Bad debts expense | 1100 |
| Advertising expense | 2500 |
| Interest expense | 500 |
| Insurance expense | 600 |
| Office supplies expense | 2025 |
| Other expenses | 7000 |
| Depreciation expense, buildings | 900 |
| Depreciation expense, office equipment | 850 |
| Total Expenses | $ 84 500 |
| **Income before taxes** | $ 25 650 |
| Income taxes | 9350 |
| **Income after taxes** | **$ 16 300** |

Classify each of the following as a **Current Asset**, **Non-current Asset**, **Current Liability**, or **Non-current Liability**:

| Accounts Head | Classification |
|---|---|
| Cash | Current Asset |
| Accounts Receivable | Curent Asset |
| Inventory | Current Asset |
| Equipment | Non-current Asset |
| Long-term debt | Non-current Liability |
| Accounts payable | current liability |
| Buildings | non-current assets |
| Notes payable (due in 6 months) | curret liability since 6 months |
| Patents | non current assets |

A patent is a type of intellectual property right that gives an inventor exclusive legal rights to their invention for a limited period (usually 20 years).